

Angular Hands-On 9

NgRx State Management

Objectives

- Understand centralized state management in Angular applications.
- Learn NgRx concepts: **Store, Actions, Reducers, Effects, Selectors.**
- Manage course enrollment state globally.
- Dispatch actions and react to state changes.
- Use selectors to read state in components.

Tasks

Task 1: Install NgRx

- Install required packages:

```
npm install @ngrx/store @ngrx/effects @ngrx/entity @ngrx/store-devtools
```

Task 2: Define Actions

- Create `course.actions.ts`.
- Add actions for:
 - Loading courses (`loadCourses`, `loadCoursesSuccess`, `loadCoursesFailure`)
 - Enrolling (`enrollCourse`)
 - Unenrolling (`unenrollCourse`)

Task 3: Implement Reducer

- Create `course.reducer.ts`.
- Define initial state:
 - `courses: []`
 - `enrolledCourseIds: []`
 - `error: null`
- Handle actions to update state accordingly.

Task 4: Create Selectors

- Create `course.selectors.ts`.
- Add selectors for:

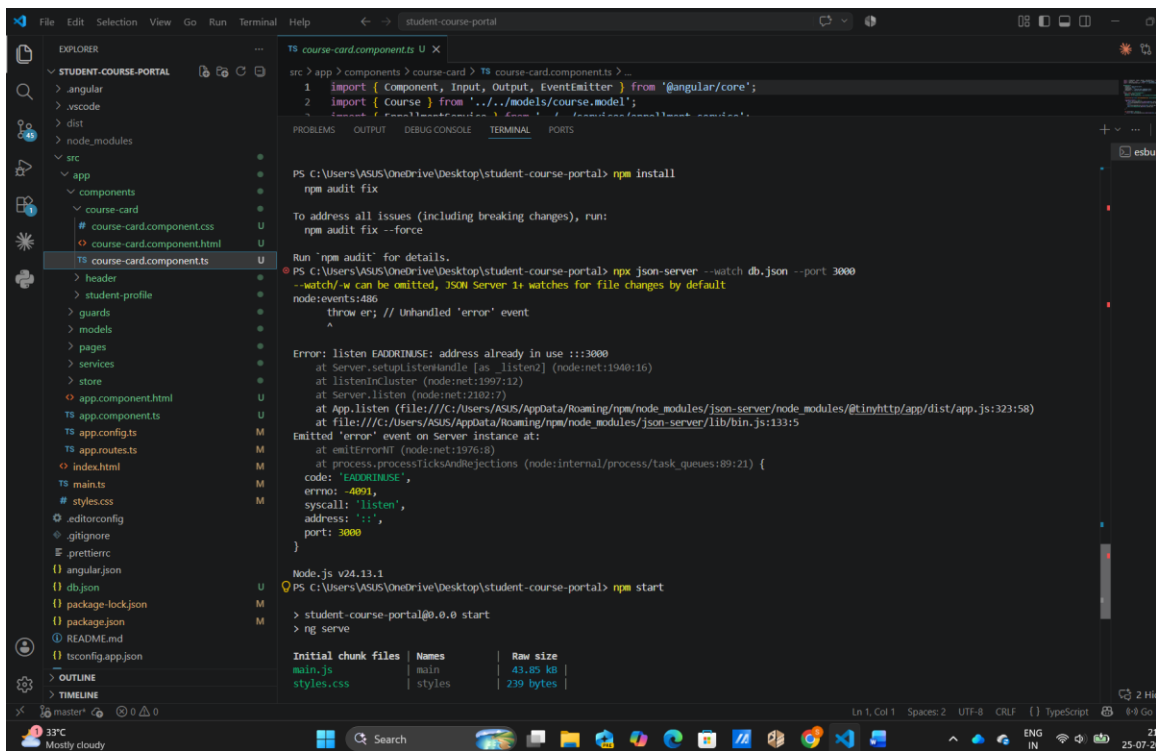
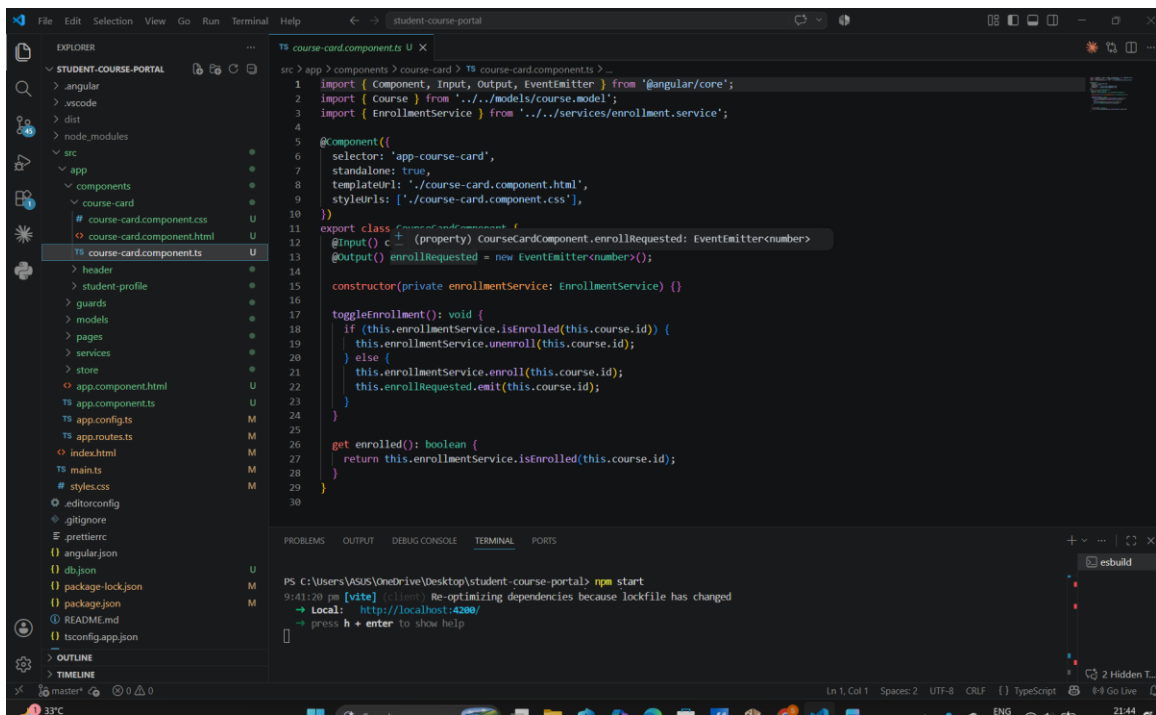
- All courses (selectAllCourses)
- Enrolled course IDs (selectEnrolledCourseIds)

Task 5: Implement Effects

- Create course.effects.ts.
- Use Actions + mergeMap to call API service.
- Dispatch loadCoursesSuccess or loadCoursesFailure.

Task 6: Connect Store to Component

- Inject Store in CourseListComponent.
- Dispatch loadCourses() in ngOnInit().
- Use selectors with store.select() to get courses and enrolled IDs.
- Bind data in template with async pipe.
- Add buttons to dispatch enrollCourse and unenrollCourse.



```

Error: listen EADDRINUSE: address already in use :::3000
    at Server.setupListenHandle [as _listen2] (node:net:1940:16)
    at listenInCluster (node:net:1997:12)
    at Server.listen (node:net:2102:7)
    at App.listen (file:///C:/Users/ASUS/AppData/Roaming/npm/node_modules/json-server/node_modules/@tinyt
    at file:///C:/Users/ASUS/AppData/Roaming/npm/node_modules/json-server/lib/bin.js:133:5
Emitted 'error' event on Server instance at:
    at emitErrorNT (node:net:1976:8)
    at process.processTicksAndRejections (node:internal/process/task_queues:89:21) {
  code: 'EADDRINUSE',
  errno: -4091,
  syscall: 'listen',
  address: '::',
  port: 3000
}

Node.js v24.13.1
PS C:\Users\ASUS\OneDrive\Desktop\student-course-portal> npm start

> student-course-portal@0.0.0 start
> ng serve

Initial chunk files | Names      | Raw size
main.js             | main       | 43.85 kB |
styles.css          | styles     | 239 bytes |

| Initial total | 44.09 kB

Application bundle generation complete. [2.237 seconds] - 2026-07-25T16:11:20.235Z

Watch mode enabled. Watching for file changes...
NOTE: Raw file sizes do not reflect development server per-request transformations.
9:41:20 pm [vite] (client) Re-optimizing dependencies because lockfile has changed
→ Local: http://localhost:4200/
→ press h + enter to show help

```